FEEDBACK INTELLIGENCE DASHBOARD

Turning Student Feedback into Real-Time Mentor Intelligence

Futurense Institute · Analytics Platform · March 2026

Internal Presentation Report

Table of Contents

01

What Is This Platform?

An overview of the system and what it delivers

The Feedback Intelligence Dashboard is a purpose-built analytics platform for Futurense Institute. It automatically collects student feedback after every session, processes it through a proprietary scoring engine, and presents the results as an interactive, real-time command centre for academic leadership.

In simple terms: every time a student rates a session, that rating flows automatically into this dashboard within minutes — no manual data entry, no spreadsheet juggling, no waiting for weekly reports. Leadership can see exactly how every mentor is performing, right now.

15,000	8	3	6	300	Real-time
Feedback Records	Mentors Tracked	Programs Covered	Cohorts Monitored	Students	Live Data Sync

What makes it different from a simple average?

Most feedback systems stop at 'average rating'. This platform goes further. It asks: *Is this mentor consistent week after week? Do they have enough data for us to trust this score? Are they trending upward or declining?* The answers combine into a single, trustworthy performance score — the Mentor Performance Index.

02

The Problem We Solved

Why this platform was needed

Before this platform, the institution faced four critical gaps:

■ No pattern detection

Feedback was collected via Google Forms but reviewed manually. Nobody could tell if a mentor's ratings were declining week over week, or if a single bad week was just noise.

■■ Always looking backward

Reports were generated periodically, meaning issues were discovered weeks after they could have been addressed. By the time a pattern was visible, it was already entrenched.

■ Average ratings hide the truth

A mentor with a 3.8 average could be consistently average, or alternating between 4.8 and 2.8. These are completely different situations — but a simple average treats them the same.

■ No early warning system

There was no mechanism to automatically flag mentors who needed attention before students became disengaged. Intervention was reactive, not proactive.

> *"The goal was not just to collect feedback — it was to make that feedback speak. To turn hundreds of ratings per week into a clear signal that leadership could act on immediately."*

03 How It Works — The Big Picture

The journey from student tap to leadership insight

The platform operates as a fully automated pipeline. A student submits feedback, and within minutes — with no human intervention — that feedback has been processed, scored, and is visible on the dashboard:

1 Student submits feedback

After each session, students complete a Google Form: a 1–5 star rating, a structured response (Excellent / Good / Average / Poor), and an optional free-text comment.

2 Automatic sync to database

The backend polls Google Sheets every 60 seconds. New responses are validated, matched to the correct program, cohort, mentor, and session, then stored in the database. Duplicates are automatically prevented.

3 MPI engine processes the data

The Mentor Performance Index algorithm runs against all new data. It calculates quality, consistency, reliability, and trend scores for every mentor, then classifies each into a performance category.

4 AI generates executive briefs

Google Gemini 2.5 Flash reads each mentor's metrics alongside actual student comments to produce a written narrative brief — strengths, concerns, and a strategic recommendation.

5 Dashboard updates automatically

The React frontend refreshes silently every 30 seconds. Charts, scores, leaderboards, and alerts all update without the user needing to do anything.

04

The Mentor Performance Index (MPI)

Our core innovation — a score you can trust

The Mentor Performance Index is the intellectual core of this platform. It is a single number between 0 and 1 that summarises a mentor's overall performance — not just how high their ratings are, but how *trustworthy*, *consistent*, and *improving* that performance is.

Why not just use the average rating?

Average ratings are easy to understand but dangerously misleading. Consider two mentors, both with a 3.8 average rating across 10 sessions:

Mentor	Average	Weekly Pattern	What it means
Mentor A	3.8	4.0, 3.9, 3.8, 3.7, 3.9 ...	Consistent, stable, predictable
Mentor B	3.8	1.5, 5.0, 1.8, 4.9, 2.0 ...	Highly volatile — students experience a lottery

The MPI correctly scores Mentor A higher than Mentor B, because consistency matters. Students of Mentor B cannot predict whether today will be a great session or a terrible one — and that unpredictability erodes trust.

$$\text{MPI} = 0.45 \times \text{Quality} + 0.20 \times \text{Consistency} + 0.20 \times \text{Reliability} + 0.15 \times \text{Trend}$$

A weighted composite of four independently calculated scores, each between 0 and 1

Each of the four components captures a different dimension of what it means to be a good mentor. The next section explains each one in detail, including the exact formula and the reasoning behind it.

05 The Four Pillars of MPI

How each component is calculated and why

1 Quality (Q) — Weight: 45%

$$Q = (\text{Average Rating} - 1) / 4$$

What it measures: Normalises the average rating from the 1-5 scale onto a 0-1 scale.

Why this weight: Quality carries the highest weight (45%) because raw performance is the most direct measure of effectiveness.

Example: A mentor with a 4.6 average scores Q = (4.6-1)/4 = 0.90. A 3.0 average gives Q = 0.50.

Why this formula: Subtracting 1 removes the floor of the scale — a rating of 1 should map to zero quality, not 0.2. Dividing by 4 (the range) ensures the score uses the full 0-1 range instead of being compressed between 0.2 and 1.0.

2 Consistency (C) — Weight: 20%

$$C = \max(0,\ 1 - \text{Standard Deviation} / 2)$$

What it measures: Penalises variance in ratings. A mentor who scores around 4.0 every week rates higher than one who alternates between 2.0 and 5.0.

Why this weight: Consistency carries 20% because it directly affects the student experience. Unpredictable quality erodes trust.

Example: Std dev 0.4 gives C = 0.80 (stable). Std dev 1.8 gives C = 0.10 (highly volatile).

Why this formula: Dividing by 2 calibrates the penalty for a 1-5 scale. A standard deviation of 2 represents extreme volatility (swinging near-1 to near-5 regularly), mapping to C=0. The max(0,...) floor prevents impossible negative values.

3 Reliability (R) — Weight: 20%

$$R = 1 - e \wedge (-\text{Feedback Count} / 1000)$$

What it measures: Measures how much we can trust the score. A mentor with 50 ratings is less statistically reliable than one with 2,000.

Why this weight: Reliability carries 20% because a score from 30 data points should not carry the same weight as one from 3,000.

Example: 500 feedbacks: R = 0.39. 2,000 feedbacks: R = 0.86. 5,000 feedbacks: R = 0.99.

Why this formula: The exponential function grows quickly at first (small samples improve confidence fast) and then levels off naturally. Dividing by 1,000 calibrates the curve to our data scale — at ~1,875 feedbacks per mentor, R reaches approximately 0.85.

4 Trend (T) — Weight: 15%

$$T = \text{clamp}(\ \text{Weekly Slope} \times 6,\ -1,\ +1\ )$$

What it measures: Captures whether a mentor is improving or declining. Calculated from linear regression over the last four weeks of sessions.

Why this weight: Trend carries 15% — meaningful but not dominant. A mentor should not be heavily penalised for a temporary dip.

Example: Slope +0.10/week gives T = +0.60 (improving). Slope -0.12/week gives T = -0.72 (concerning decline).

Why this formula: Multiplying by 6 converts the tiny weekly slope (typically 0.05-0.15) into a meaningful -1 to +1 range. Clamping prevents outlier weeks from producing extreme values. Using the last 4 weeks keeps the signal recent.

06 Mentor Categories — Explained

What each classification means and what to do

Every mentor is automatically classified into one of five categories based on their MPI score and statistical position relative to the institution. These categories drive the visual alerts, risk indicators, and AI brief tone.

■ Top Performer

Condition: MPI >= 0.72 AND Z-score > 0

Action: Benchmark and celebrate. Consider as peer mentor for others.

■ Stable

Condition: 0.65 <= MPI < 0.72

Action: Healthy baseline. Monitor and maintain.

■■ Needs Attention

Condition: 0.55 <= MPI < 0.65

Action: Schedule a feedback conversation. Identify specific pain points.

■ High Volatility

Condition: Consistency < 0.50 AND avg > 3.5

Action: Investigate root cause. Good average hiding dangerous inconsistency.

■ At Risk

Condition: MPI < 0.55 OR Z-score < -1.5

Action: Immediate intervention required. Do not wait for next review cycle.

Note on the Z-Score: The Z-score measures how far a mentor's average rating is from the institutional average, in standard deviation units. A Z-score of +1 means one standard deviation above average — statistically distinguished performance. Below -1.5 means performance is severely below the institutional norm, triggering the At Risk flag even if the raw MPI is borderline.

07

Why These Numbers?

The calibration decisions and reasoning behind them

Every threshold and weight in the MPI formula was a deliberate choice, not a guess. Here is the reasoning behind the most important decisions:

Why 45% for Quality?

Quality is the most direct signal of a mentor's impact on students. If ratings are consistently high, that fundamentally reflects effective teaching. The 45% weight ensures raw performance dominates the score, while leaving room for the three contextual factors to modulate it.

Why equal weight (20%) for Consistency and Reliability?

Both factors address the question of 'can we trust this score?'. Consistency checks if the performance is real and stable. Reliability checks if we have enough data to be confident. Neither is more important than the other — both must be true for a score to be truly trustworthy.

Why 15% for Trend?

Trend is forward-looking — it tells you where a mentor is headed. It deserves meaningful weight but should not override current performance. A mentor who is excellent but had two slightly lower weeks recently should not drop categories because of it.

Why is the Top Performer threshold 0.72, not 0.75?

This was calibrated against actual data. The Reliability formula R = 1 - e^(-count/1000) is exponential. At our average data volume of ~1,875 feedbacks per mentor, R reaches approximately 0.85. This means even a mentor with perfect quality and consistency cannot mathematically reach an MPI of 0.75 at current volumes. The threshold was adjusted to 0.72 to ensure the category is achievable and meaningful — it still requires both a positive Z-score AND an MPI above 0.72, so it remains genuinely selective.

Why is the At Risk threshold 0.55?

Below 0.55, a mentor's performance represents a meaningful failure across multiple dimensions. For context: a mentor with a 3.0 average rating (mediocre), moderate consistency, and sufficient data would score around 0.56 — just above the threshold. An MPI below 0.55 implies poor quality, severe inconsistency, low data confidence, or some combination of all three.

Why 1,000 as the divisor in Reliability?

The divisor calibrates where the confidence curve peaks. At 1,000 feedbacks, R = 0.63 — reasonable but not high. At 2,000 it reaches 0.86, at 3,000 it reaches 0.95. Given that our average mentor has ~1,875 feedbacks, this calibration means no mentor is penalised excessively for normal data volumes, while also not giving false confidence to mentors with only a few dozen responses.

08 What You See — Dashboard Features

A tour of the seven pages and what each reveals

■■ Overview — Institution View

The first screen. Shows the entire institution's health at a glance: total feedback volume, average rating, mentor category distribution, top performing programs. Click a program to drill into its cohorts. Click a cohort for its full dashboard. Refreshes silently every 30 seconds.

- Mentor Health Bar — proportional colour bar showing the split across all five categories
- Top 3 Mentor Spotlight — gold, silver, bronze cards for the institution's highest MPI scorers
- Sentiment Distribution — donut chart of Positive / Neutral / Negative feedback proportions

■■■ Mentor Intelligence

A full ranked table of all 8 mentors sorted by MPI. For each mentor, animated score bars show Quality, Consistency, and Reliability, alongside their category badge and MPI score. Click any row to open that mentor's full individual dashboard.

- Risk Quadrant — scatter plot positioning each mentor by MPI vs average rating
- Compare shortcut — button to jump directly to the Mentor Comparison page

■ Cohort Analytics

All six cohorts grouped by program. Each card shows the cohort's average rating, feedback count, number of mentors, and a visual health indicator. Cards with at-risk mentors are flagged automatically.

- Per-cohort rating bar — animated progress bar showing avg rating relative to 5.0
- At-risk badge — appears on any cohort where a mentor has been classified At Risk

■ Session Trends

Two charts: institution-wide average per week (the single most important leading indicator), and individual mentor lines with toggle buttons to show or hide specific mentors for comparison.

- Weekly summary table — each week's institution average, best performer, and lowest scorer
- Mentor toggles — click any mentor name to show/hide their trend line on the chart

■ AI Insights

Rule-based intelligence generated for all mentors simultaneously. Five category summary cards at the top act as clickable filters. Each mentor's insight card explains their situation in plain language.

- Category filter — instantly show only Top Performers, At Risk, or any other category
- Count summary — how many mentors fall into each category at a glance

■ Mentor Detail Page

The deepest view. Accessed by clicking any mentor anywhere in the app. Shows all four score breakdowns, all charts, recent student comments filtered by rating tier, and the full AI-generated executive brief.

- Comment Explorer — filter student comments by Excellent / Good / Average / Poor
- AI Executive Brief — full narrative covering strengths, concerns, student voice, and recommendation
- Export buttons — download data as CSV, or print the page as a PDF report

■■ Mentor Comparison

Select up to four mentors to compare side by side. A radar chart overlays all four across five dimensions simultaneously.

- Radar chart — visual comparison of Quality, Consistency, Reliability, Trend, and Avg Rating
- Metrics table — best value in each row is highlighted with an arrow indicator

09 AI-Powered Intelligence

How the platform reads between the numbers

Numbers tell you *what* happened. The AI tells you *why it matters* and *what to do about it*.

Each mentor's detail page includes an executive brief generated by Google Gemini 2.5 Flash. The brief is not a template with values filled in. It reads the mentor's actual MPI breakdown, their position relative to institutional benchmarks, and up to 50 of their most recent student comments — then writes an original analytical narrative.

The brief has six structured sections:

Executive Snapshot	A one-paragraph summary of the mentor's current performance in context.
Strength Signals	Specific metrics and comment themes that indicate what the mentor does well.
Improvement Signals	Areas of concern, backed by data — not vague suggestions.
Student Voice	Recurring patterns extracted from the actual words students used in their comments.
Momentum & Outlook	Whether the mentor is improving, stable, or declining, and how confident we are.
Strategic Recommendation	One concrete, actionable next step for the institution to take.

The AI brief is generated once per mentor and cached — the first click may take a few seconds, but subsequent visits are instant. This avoids unnecessary API calls while ensuring the brief is always available when leadership needs it.

10 Live Data & Real-Time Sync

How the platform stays current automatically

One of the most important design decisions was making the system truly live. Data is not refreshed by clicking a button or running a report — it flows continuously.

60 sec	How often the backend polls Google Sheets for new form responses
30 sec	How often the frontend silently refreshes all charts and scores without any page reload
~1 min	Approximate time from a student submitting feedback to it appearing on the dashboard
Zero	Manual steps required — no refresh button, no scheduled reports, no CSV uploads
■K	Command palette shortcut to instantly jump to any page or mentor from anywhere in the app

The sidebar always shows a live 'Last synced X seconds ago' indicator that counts up in real time. A pulsing green dot confirms the connection is active. When data refreshes, a subtle notification appears so the user knows the view has updated without any disruption to their current view.

11 Impact & Key Numbers

What the platform has delivered

15,000
Feedback Records Processed

8
Mentors Scored and Ranked

5
Performance Categories

7
Dashboard Pages

4
MPI Dimensions Measured

AI
Executive Brief per Mentor

30s
Auto-Refresh Interval

■K
Command Palette Navigation

What changed for the institution:

From weekly reports to live data

Feedback is visible within minutes of submission, not days after the fact.

From averages to multi-dimensional scores

The MPI captures quality, consistency, reliability, and trajectory simultaneously.

From reactive to proactive

At Risk and Needs Attention flags alert leadership before problems become crises.

From data to narrative

AI briefs translate numbers into plain-language insights that any stakeholder can understand.

From static to interactive

Drill down from institution level to individual mentor level in three clicks.

12 Summary & Strategic Value

What this platform means for Futurense Institute

The Feedback Intelligence Dashboard is more than a reporting tool. It is a strategic asset that transforms how the institution understands and acts on the most direct signal available to academic leadership: what students experience in every session, every week.

Three things this platform enables that were not possible before:

Early intervention

At Risk mentors are identified automatically, typically weeks before the pattern would become visible in a periodic review. This is the difference between a conversation and a crisis.

Evidence-based development

When a mentor needs support, the AI brief provides specific, data-backed talking points. The conversation starts from 'here is what the data shows' rather than 'we have heard some feedback'.

Institutional benchmarking

The Z-score and comparison tools allow the institution to understand not just individual performance, but how the entire mentor pool is distributed relative to the institution's own standards.

This platform was designed around one principle: feedback only has value if it leads to action. Every feature — from the MPI formula to the AI brief to the live sync — exists to close the gap between a student's experience and a leader's decision.

Platform	Feedback Intelligence Dashboard
Institution	Futurense Institute
Core Metric	Mentor Performance Index (MPI) — 4-dimensional composite score
Data Volume	15,000 feedback records across 3 programs, 6 cohorts, 8 mentors
Refresh Rate	60-second backend sync; 30-second frontend refresh
AI Integration	Google Gemini 2.5 Flash — executive briefs with student comment analysis
Dashboard Pages	7 pages: Overview, Mentor Intelligence, Cohort Analytics, Trends, AI Insights, Mentor Detail, Comparison
Categories	Top Performer / Stable / Needs Attention / High Volatility / At Risk